Dynamical Threshold Theory of Sleep: A Network Prototype for Recovery Phase Transitions

Keiji Yoshimura

Independent Researcher

April 16, 2026

Abstract

Sleep is widely associated with homeostatic recovery, local use-dependent regulation, and state-dependent reorganization of network interactions, yet these motifs are rarely integrated within a single dynamical framework. Here we propose the *Dynamical Threshold Theory of Sleep*, a prototype theory in which a multicellular biological system is represented as a network of adaptively coupled oscillatory elements endowed with local homeostatic load, local sleep propensity, and selectively protected interactions. In this framework, wakefulness is modeled as a noisy adaptive synchronization regime under continued perturbation, whereas sleep emerges when the joint accumulation of local load and degradation of local coherence drive the system across a collective dynamical threshold into a sleep-dominant renormalization phase. To make this transition mathematically well posed, the sleep-dominant subsystem is formulated as a bounded gradient flow of a reduced free energy, thereby avoiding the unbounded-coupling pathology of naive adaptive phase-oscillator models. Numerical proof-of-concept simulations on hierarchical modular small-world networks with localized filtered-Poisson drive reproduced three central predictions of the theory: spatially heterogeneous accumulation of homeostatic burden, rapid growth of the mean sleep propensity into a sleep-dominant regime, and bounded global downscaling of mean coupling strength toward finite fixed points. Additional analyses further showed that the reduced free energy relaxes after an early transient overshoot toward a bounded plateau, and that edges with stronger pre-sleep protection are preferentially preserved during sleep, consistent with selective down-selection rather than uniform weakening. These results support the present framework as a mathematically closed prototype for treating sleep-like transitions as recovery phase dynamics in adaptive multicellular systems.

1 Introduction

Sleep is among the most widespread and deeply conserved phenomena in biology, yet its theoretical status remains surprisingly fragmented. At the descriptive level, sleep is associated with reduced responsiveness to external stimuli, homeostatic rebound after deprivation, characteristic changes in large-scale activity, and reversible alternation with wakefulness. At the mechanistic level, however, different explanatory traditions emphasize different aspects of the phenomenon: sleep has been linked to homeostatic recovery, metabolic and cellular restoration, local use-dependent regulation, synaptic renormalization, and circadian control [1–3, 8, 9]. These perspectives are individually productive, but they are often formulated at different levels of description and rarely expressed within a single network-dynamical formalism.

One influential line of work proposes that wakefulness carries a cumulative cost because plastic interactions strengthened during waking activity increase energetic demand, reduce signal-to-noise ratios, and progressively saturate the capacity for further learning. In this view, sleep serves to renormalize net synaptic strength while selectively preserving functionally important traces, thereby restoring an efficient operating regime [1–4]. A second line of work emphasizes that sleep need not begin as an all-or-none global switch: sleep-like events and OFF-period-like phenomena can arise locally, can coexist with behavioral wakefulness, and can increase with prior wake duration, implying that sleep regulation is at least partly local and use dependent [5–7]. A third line of work, formalized most prominently in the two-process tradition, frames sleep timing as the interaction between a history-dependent homeostatic process and a circadian process that gates when sleep pressure is most likely to be expressed [8, 9]. Taken together, these lines of evidence suggest that sleep is best understood not as a purely top-down binary switch, but as an emergent collective phenomenon produced by the interaction of local load accumulation, network coherence, adaptive connectivity, and temporally structured regulation.

Despite this conceptual convergence, a mathematically closed framework connecting these ingredients remains underdeveloped. Synchronization models such as the Kuramoto family provide a tractable language for collective order and disorder in systems of coupled oscillators, and they have been fruitfully extended to complex networks, including small-world topologies that support rapid global coordination while retaining local structure [10, 11]. What is missing is a prototype theory that can simultaneously represent stochastic perturbation during wakefulness, local accumulation of burden, the gradual emergence of local sleep propensity, the transition to a sleep-dominant regime, and the selective renormalization of coupling structure.

The present work introduces such a prototype in the form of the *Dynamical Threshold Theory of Sleep* (DTTS). In DTTS, a multicellular system is represented as a network of adaptively coupled oscillatory units whose phases evolve under noisy perturbation, while each node also carries a local homeostatic load and a local sleep gate. Wakefulness is modeled as an adaptive synchronization regime under continued perturbation and use-dependent drive. Sleep is defined by a dynamical threshold mechanism: as local coherence degrades and local load accumulates, sleep propensity rises across the network, and once enough of the system crosses threshold, the dynamics enter a sleep-dominant renormalization phase.

2 Model

2.1 Network and Phase Dynamics

We represent a multicellular biological system as a network of N coupled oscillatory nodes on an undirected graph $G = (V, E)$ with adjacency matrix $A_{ij} \in \{0, 1\}$. The phase evolution of each node is governed by:

$$\frac{d\theta_i}{dt} = \omega_i + \sum_{j=1}^{N} A_{ij} K_{ij} \sin(\theta_j - \theta_i) + \sigma_i (1 - q_i) \xi_i(t), \tag{1}$$

where ω_i is the intrinsic frequency, σ_i is the susceptibility to external perturbation, and $\xi_i(t)$ represents independent Gaussian white noise. The local coherence is quantified by $R_i(t) = |\frac{1}{d_i} \sum_{j=1}^{N} A_{ij} e^{i(\theta_j - \theta_i)}|$.

2.2 Local Load and Sleep Propensity

The local homeostatic load $s_i(t)$ evolves as:

$$\frac{ds_i}{dt} = a(1 - R_i) + \chi u_i(t) - b q_i s_i, \tag{2}$$

where $u_i(t)$ is the spatially localized use-dependent drive. The local sleep gate $q_i(t)$ is given by:

$$\tau_q \frac{dq_i}{dt} = -q_i + \Sigma\Big(\beta_s (s_i - s_c) - \beta_c C(t) - \beta_R (R_i - R_c)\Big), \tag{3}$$

where $C(t)$ represents the circadian rhythm and $\Sigma(x)$ is the sigmoid function.

2.3 Selective Protection and Coupling Updates

The adaptive coupling weights $K_{ij}(t)$ follow:

$$\frac{dK_{ij}}{dt} = (1 - \bar{q}_{ij})\Big[\eta_W P_{ij}(K_{\max} - K_{ij}) - \delta_W K_{ij}\Big] - \bar{q}_{ij}\eta_K\Big(\mu K_{ij} - \kappa P_{ij}\Big), \tag{4}$$

where $P_{ij}(t)$ is the protection field representing coherence history. During sleep ($\bar{q}_{ij} \approx 1$), couplings relax toward a protection-dependent fixed point $K^*_{ij} = (\kappa/\mu)P_{ij}$.

3 Analytical Properties

3.1 Reduced Free-Energy Formulation

The sleep-dominant regime can be formulated as a gradient flow of a reduced free energy $\mathcal{F}_s$:

$$\mathcal{F}_s(K, s \mid P) = \sum_{i<j} A_{ij}\left(\frac{\mu}{2}K_{ij}^2 - \kappa P_{ij}K_{ij}\right) + \frac{\nu}{2}\sum_{i=1}^{N} s_i^2. \tag{5}$$

In the limit $q_i \to 1$, the dynamics satisfy $\dot{K}_{ij} \propto -\frac{\partial \mathcal{F}_s}{\partial K_{ij}}$ and $\dot{s}_i \propto -\frac{\partial \mathcal{F}_s}{\partial s_i}$, ensuring $\frac{d\mathcal{F}_s}{dt} \leq 0$.

4 Numerical Results

Numerical simulations on a hierarchical modular small-world network reproduced the transition to a sleep-dominant state.

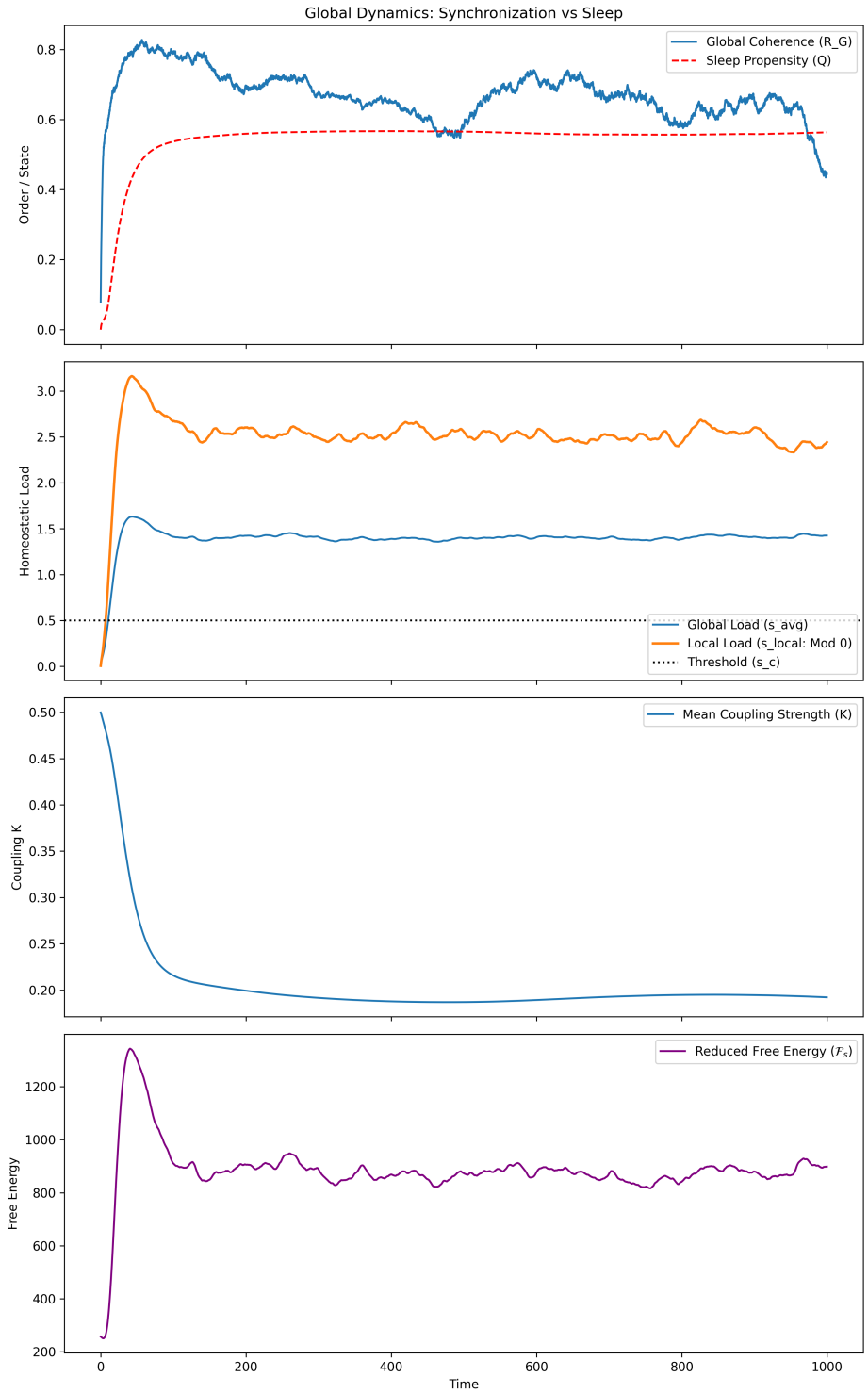

Figure 1: Summary of the DTTS simulation. Top to bottom: global coherence R_G and mean sleep propensity Q; homeostatic load s; mean coupling strength $\bar{K}$; and reduced free energy $\mathcal{F}_s$.

The simulation confirmed that edges with a high pre-sleep protection field P_{ij} are pref-

erentially preserved, while unstructured edges are aggressively down-selected.

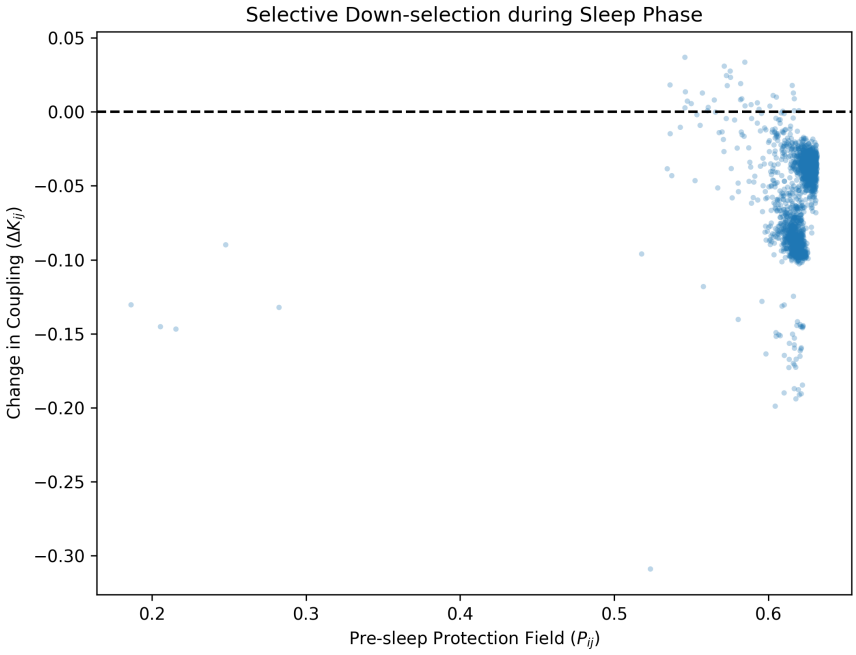

Figure 2: Scatter plot of coupling change ΔK_{ij} versus pre-sleep protection field P_{ij}.

5 Conclusion

The Dynamical Threshold Theory of Sleep provides a mathematically closed framework linking local burden, synchronization degradation, and adaptive network renormalization. This prototype establishes a rigorous foundation for conceptualizing sleep as an emergent recovery-phase phenomenon in complex adaptive systems.

Acknowledgments

The author conceptually designed the study and formulated the core hypotheses. The author acknowledges the use of large language models as computational and theoretical assistants to formalize the mathematical framework and generate the numerical simulation code under the author's direct supervision and validation.

References

[1] Tononi, G., & Cirelli, C. (2003). Sleep and synaptic homeostasis: a hypothesis. *Brain Research Bulletin*, 62(2), 143–150.

[2] Tononi, G., & Cirelli, C. (2006). Sleep function and synaptic homeostasis. *Sleep Medicine Reviews*, 10(1), 49–62.

[3] Cirelli, C., & Tononi, G. (2015). Sleep and Synaptic Homeostasis. *Sleep*, 38(10), 1615–1624.

[4] Tononi, G., & Cirelli, C. (2014). Sleep and the Price of Plasticity. *Neuron*, 81(1), 12–34.

[5] Vyazovskiy, V. V., et al. (2011). Local sleep in awake rats. *Nature*, 472(7344), 443–447.

[6] Krueger, J. M., et al. (2019). Local Sleep. *Sleep Medicine Reviews*, 43, 14–21.

[7] Siclari, F., et al. (2018). Local aspects of sleep and wakefulness. *Current Opinion in Neurobiology*, 44, 222–227.

[8] Borbely, A. A., et al. (2022). The two-process model of sleep regulation. *Journal of Sleep Research*, 31(6), e13598.

[9] Deboer, T. (2018). Sleep homeostasis and the circadian clock. *Neurobiology of Sleep and Circadian Rhythms*, 5, 68–77.

[10] Acebron, J. A., et al. (2005). The Kuramoto model. *Reviews of Modern Physics*, 77(1), 137–185.

[11] Hong, H., et al. (2002). Synchronization on small-world networks. *Physical Review E*, 65(2), 026139.